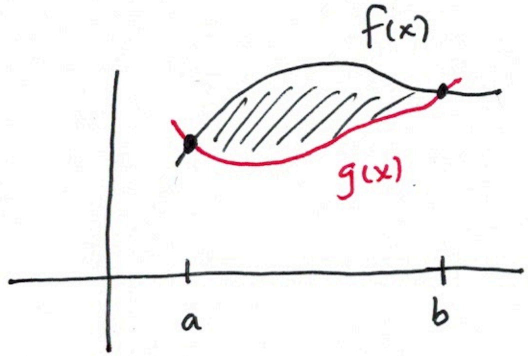
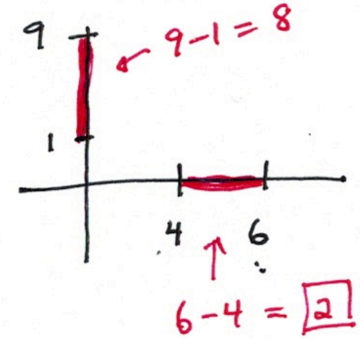
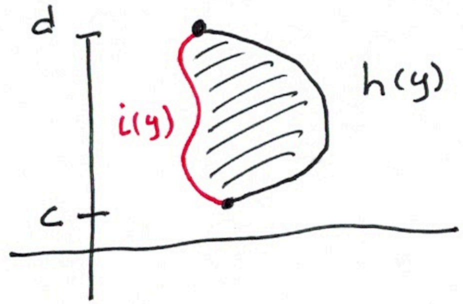
Section 6.1 Area Between Curves





top - bottom

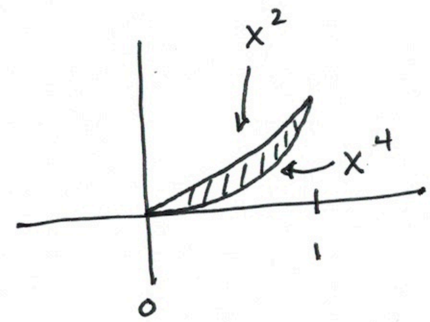
$$\int_a^b (f(x) - g(x))\,dx$$



$$\int_c^d (h(y) - i(y))\,dy$$

Ex ① Find the finite area between

$y = x^2$ and $y = x^4$



$$2 \cdot \int_0^1 (x^2 - x^4)\, dx$$

$$2 \cdot \left(\frac{x^3}{3} - \frac{x^5}{5} \right) \Big|_0^1$$

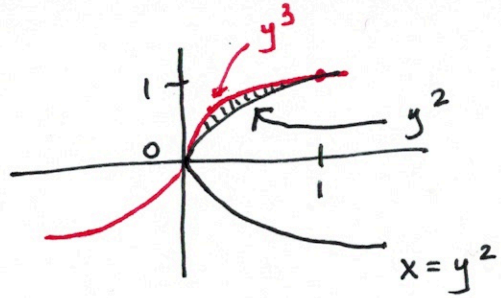
$$2 \cdot \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$2 \cdot \left(\frac{2}{15} \right)$$

$$\frac{4}{15}$$

Ex ② $x = y^3$

$x = y^2$



$$y^3 = y^2$$

$$y^3 - y^2 = 0$$

$$y^2(y-1) = 0$$

$$y = 0 \qquad y = 1$$

$$\int_0^1 (y^2 - y^3)\, dy$$

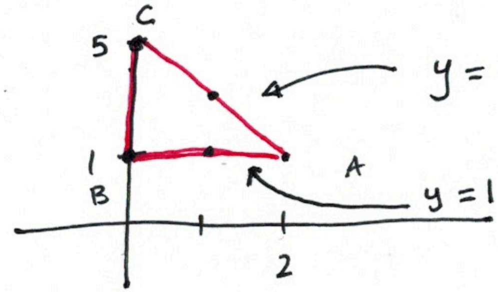
$$\left(\frac{y^3}{3} - \frac{y^4}{4}\right)\Bigg|_0^1$$

$$\frac{1}{3} - \frac{1}{4}$$

$$\boxed{\frac{1}{12}}$$

Ex ③ Set up integral for area of ΔABC

$A(2,1) \quad B(0,1) \quad C(0,5)$



$A = \frac{1}{2}(2)(4)$

$A = 4$ ✓

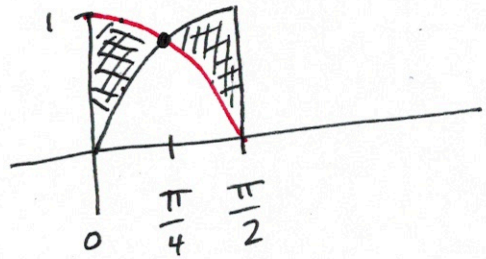
$$\int_0^2 (-2x + 5 - 1)\,dx$$

$$(-x^2 + 4x)\Big|_0^2$$

$$-4 + 8 - 0$$

$$\boxed{4}$$ ✓

④ $y = \sin x, \quad y = \cos x, \quad [0, \frac{\pi}{2}]$



$$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}$$

$$2 \cdot \int_0^{\frac{\pi}{4}} (\cos x - \sin x)\, dx$$

$$2 \cdot \left[\sin x + \cos x \right]_0^{\frac{\pi}{4}} \rightarrow 2\left[\sin\frac{\pi}{4} + \cos\frac{\pi}{4} - \sin 0 - \cos 0 \right]$$

$$= .828$$